Rutgers-The State University
Master's Comprehensive Examination

Part I — **Theory** — December 1, 1984

Instructions:
1. Open book and open notes
2. Do four out of the following six problems
3. Use a separate blue book for this section of the exam
4. Indicate which questions you wish graded

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1. A box contains three \$1.00 bills and two \$2.00 bills. A coin is tossed. If a head appears one bill is selected, at random, from the box. If a tail appears, two bills are selected, at random, from the box. Let X = the total money taken from the box. Find

   a) $E\{X\}$.
   b) $Var(X)$.
   c) P(coin toss resulted in heads | X = \$2.00).

2. a) Let $X_i$, $i=1\ldots n_1$ have distribution $N(\mu_{X_1}\ \sigma_X^2)$

   Let $Y_j$, $j=1\ldots n_2$ have distribution $N(\mu_{Y_1}\ \sigma_Y^2)$

   all random variables are independent. Find $P[\overline{X} > \overline{Y}]$, Give the numerical value of this probability for $n_1=10$, $n_2=15$, $\mu_X = \mu_Y$ and $\sigma_X^2 = \sigma_Y^2 = 6$.

   b) Let $X_1$, $X_2$ be independent r.v. distributed as negative exponential, with parameter $\theta = 1$, derive the density of $\sqrt{X_1 + X_2}$

3. Let $X_1,\ldots,X_{10}$ be i.i.d. uniform random variables on $(0,\theta)$. Find an upper confidence bound for $\theta$ with confidence coefficient .95

4. Let $X_1,\ldots,X_n$ be i.i.d. with a density

$$f(x) = \theta\, e^{-\theta(x-\phi)},\ x > \phi\ ,\ \theta > 0,\ \phi \text{ known.}$$

   (i) Consider testing $H_0$: $\theta < \theta_0$ vs $H_1$: $\theta > \theta$ at level $\alpha$ let the critical region be $\overline{X} < c_\alpha$ for some constant $c_\alpha$, show this to be a UMP level $\alpha$ test.

   (ii) For large n using central limit theorem find an approximate value for the critical region for $\alpha = .05$.

   (iii) Using normal approximation find the smallest value of n required to have a power at least .95 for all alternatives $\theta > 1{\cdot}1\ \theta_0$ when the level of test is .01 .

5. Consider a random sample of size n from a distribution with density $f_X(x|\underset{\sim}{\theta}) = \theta_1\theta_2^{\theta_1}/x^{\theta_1+1}$, $x > \theta_2$, $\theta_1, \theta_2 > 0$. For $\theta_1$ known, find a sufficient statistic, the maximum likelihood estimator, and an unbiased estimator of $\theta_2$.

6. Let $X_1, \ldots, X_n$ be i.i.d $N(\theta,1)$
   (i) Find the m.ℓ.e. for $\theta^2$. Is it unbiased? What is its asymptotic distribution?
   (ii) Find an unbiased estimator for $\theta^2$.
   (iii) Compare the m.s.e's for the estimators you find in (i) and (ii).

Rutgers-The State University
Master's Comprehensive Examination

Part II

Applied

December 1, 1984

Instructions:
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2. Do four out of the following five problems
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1. In 1983 of the 2000 Freshmen entering Rutgers College 10% had SAT scores over 700 and 35% had SAT scores over 600. In 1982 comparable figures were 12% and 36%. On the basis of this information test whether the quality of incoming freshmen is changing. Describe in quantitative terms any changes you find.

2. To test a method for the quantitative analysis of calcium oxide, known quantities X in milligrams were analyzed. With one exception (starred) all gave under estimates of the known content. Determine the regression of Y on X and whether it has a zero intecept. Test whether this starred result can be considered an outlier and at what probability.

| | | | | | |
|---|---|---|---|---|---|
| Taken X | 16.0 | 20.0 | 25.0 | 31.0 | 36.0 |
| Found | 15.6 | 19.8 | 24.5 | 31.1* | 35.5 |

3. The effect of 3 diodes on current are to be studied. As the effects might depend on temperature, observations are to be taken on each diode at each of 3 temperatures. The temperature is set and each diode is tested in random order.

The results are

| | diode 1 | diode 2 | diode 3 |
|---|---|---|---|
| Temp 1 | 4.0 | 4.8 | 5.0 |
| 2 | 4.4 | 5.0 | 5.2 |
| 3 | 4.8 | 4.6 | 5.2 |

a) What type of design is this
b) Calculate the ANOVA table, what model are you fitting? What do you conclude? Find the EMS.
c) If it was known from previous experiments that the variance of the measurements is approximately .1 what would you conclude?

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Applied

4. The following data is sales of a liquid chemical.
Sales for 200 days
(Thousands of gallons)

| | # of days |
|---|---|
| less than 34 | 0 |
| 34.0 and under 37.0 | 33 |
| 37.0 and under 40.0 | 78 |
| 40.0 and under 43.0 | 78 |
| 43.0 and under 46.0 | 11 |
| 46.0 or more | 0 |
| | 200 |

a) At the 1% level perform a test that sales are normally distributed.
b) What other indications do you have that the data is or is not reasonably close to normal?

5. The following figures are behavior ratings before and after 10 weeks of receiving a placebo, for 10 chronic schizophrenics.

a) Use two different nonparametric tests to test the null hypothesis of no effect of this drug. Given $\alpha = 5\%$. What do you conclude? Explain any seeming contradictions.

b) Using nonparametric procedures estimate the placebo effect and give a 95% confidence interval (as close to 95% as possible)

Before 2.5, 3.41, 2.2, 1.9, 2.85, 2.64, 3.1, 4.1, 2.9, 3.03

After 1.9, 3.6, 2.5, 3.1, 2.80, 3.05, 3.32, 4.16, 3.16, 3.15

Rutgers-The State University
Master's Comprehensive Examination

December 1, 1984

## Theory

Part I

Instructions:
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1. $X_1,\ldots,X_n$ are i.i.d. from $N(2\theta + 1,1)$

   (i) Find the m.ℓ.e. of $\theta$.
   (ii) Is it unbiased? Calculate its mean square error.
   (iii) Can you find an unbiased estimator of $\theta$ which has smaller mean square error than the m.ℓ.e?

2. Let $X_1,\ldots,X_n$ be i.i.d. from a truncated Poisson distribution i.e. the probability mass function of $X_i$'s is $f(x,\theta) = \frac{1}{e^{\theta}-1}\frac{\theta^x}{x!}$, $x = 1,2,\ldots,$ $\theta > 0$. Consider the testing problem, $H_0$: $\theta = 1$, vs $H_1$: $\theta > 1$.

   (i) Find the form of the U.M.P. test (as explicitly as possible) for the above testing problem at level $\alpha$.

   (ii) For large n, using the central limit theorem find the approximate rejection region.

3. An urn contains 3 black and 2 white balls. We draw once from the urn. If the ball drawn turns out to be black we toss a coin with $p_1$=Prob (head) until one head appears. However if the ball drawn is white we toss a coin with $p_2$=Prob (head) until one head appears. Let X be the number of tosses needed to get one head.

   (i) Find the probability distribution of X.
   (ii) Calculate the mean and the variance of X.

4. Let X have a Poisson distribution with parameter $\lambda$. Let the prior distribution of $\lambda$ have the density $f(\lambda) = \lambda e^{-\lambda}$, $\lambda > 0$.

   (a) If the loss is squared error $(\delta-\lambda)^2$ find the Bayes estimator of $\lambda$.
   (b) Calculate the mean square error of the estimator in (a).
   (c) What would the Bayes estimator be if the loss function were $\frac{(\delta-\lambda)^2}{\lambda}$ ?

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5. Let the r v's x,y have the following joint density

$$f(x,y|\theta) = \begin{cases} \dfrac{2}{\theta^2} & \text{if } 0 \leqslant y \leqslant x \leqslant \theta \\ 0 & \text{otherwise} \end{cases}$$

(i) Show that $d(y) = 3y$ is an unbiased estimator of $\theta$.
(ii) Find a sufficient statistic for $\theta$.
(iii) Find an unbiased estimator of $\theta$ having smaller variance then $d(y)$.

6. Let $X_1, \ldots, X_n$ be i.i.d. with a continuous distribution symmetric about zero.

(i) How large an n is needed if we want that with probability at least .9, between 40 and 60 percent of $X_i$'s are positive? (Use Chebyshev's inequality).
(ii) Using normal approximation calculate the probability that between 40 and 60 percent of the $X_i$'s are positive for the n you got in (i).

Rutgers-The State University
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December 1, 1984

## Applied

Part II

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a) What type of design is this
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a) Use two different nonparametric tests to test the null hypothesis of no effect of this drug. Given $\alpha$ = 5%. What do you conclude? Explain any seeming contradictions.

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Rutgers-The State University
Master's Comprehensive Examination

Part I — **Theory** — March 10, 1984

Instructions:
1. Open book and open notes
2. Do four out of the following six problems
3. Use a separate blue book for this section of the exam
4. Indicate which questions you wish graded

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1. Let $X_1,\ldots,X_n$ be i.i.d Poisson with mean $\sqrt{\theta}$, $\theta > 0$.

   (a) Find the m $\ell$ e for $\theta$. Calculate its bias.

   (b) Find an unbiased estimator for $\theta$.

   (c) Compare the m.$\ell$.e and the unbiased estimator of $\theta$ in terms of the m.s.e.

2. A fair coin will be tossed twice. If no head appears coin 1 (with prob of head $p_1$) will be tossed n times. Otherwise coin 2 (with prob of head $p_2$) will be tossed n times. Let X = # of heads.

   (a) Find the probability distribution of X.

   (b) Calculate EX and Var (X).

3. Let $X_1,\ldots,X_n$ be i.i.d with the discrete density $f(x) = (1-\theta)\theta^x$, $x = 0,1,2,3,\ldots$, $0 < \theta < 1$ is unknown. Let $\theta$ have a prior density $g(\theta) = 3/2\ (1 - \theta^2)$.

   (a) Find the posterior distribution of $\theta$ given $X_1,\ldots,X_n$.

   (b) Calculate the Bayes estimator for $\theta$ under the squared error loss

4. Let $X_1,\ldots,X_n$ be i.i.d from the population with density $f(x) = \theta x^{\theta-1}$, $0<x<1$, $\theta > 0$.

   (a) Find the UMP level $\alpha$-test for $H_0$: $\theta = 1$, vs $H_1$: $\theta > 1$

   (b) For n large, calculate an approximate value of the critical region

   (c) For n = 100 calculate the approximate power of the test at $\theta = 1{\cdot}2$ for $\alpha = .01$

   [Hint: Use central limit theorem]

5. $X_1,\ldots,X_n$ are i.i.d from $f(x) = e^{-(x-\theta)}$, $x > \theta$.

   (a) Find the m.$\ell$.e for $\theta$ and calculate its distribution

   (b) Using (a) find an $(1-\alpha)100\%$ upper confidence bound for $\theta$

6. A box contains $\sqrt{\theta}$ proportion of red balls and the rest are black. Draws will be made at random with replacement until one red ball is drawn. Let X be the number of draws required.

   (a) Calculate probability distribution of X.

   (b) Find a sufficient statistics for $\theta$

   (c) Find the m.ℓ. e and an unbiased estimator for $\theta$.

Rutgers-The State University
Master's Comprehensive Examination

March 10, 1984

## Applied

Part II

Instructions:
1. Open book and open notes
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1. The following data is dose response results relating the diameter of the exhibition zone to the concentration of antibiotic per mL.

| | antibiotic levels (concentration) | | | | | |
|---|---|---|---|---|---|---|
| | 40 | 60 | 80 | 100 | 120 | 140 |
| response | 0.6 | 1.5 | 2.6 | 2.8 | 3.2 | 3.5 |

1. Fit an appropriate linear regression (use log concentration)
2. Find 95% C.I for another observation to be taken at 120 log units.
3. test the hypotheses that the slope = 5.0

2. a. A supermarket sells brands of coffee in 1-pound cans. In a sample of selling days, sales by brand are shown below. Are sales uniformly distributed by brand? Test at $\alpha = 0.05$. State the conclusion and evidence leading to it.

| | Brand | | |
|---|---|---|---|
| | C | B | A |
| # of cans sold | 175 | 235 | 180 |

b. The following table consists of times taken by 254 college freshmen to complete a standard math exam.

| minutes to complete test | # of cases |
|---|---|
| < 22 | 10 |
| 22-30 | 61 |
| 30-38 | 122 |
| 38-46 | 57 |
| 47-55 | 4 |
| > 55 | 0 |

Perform a goodness of fit test for normality.

3. a. In a random sample of 300 students at Rutgers 120 expressed a preference for candidate G. Hart. Using a confidence coefficient of 0.9 construct the confidence interval estimate of the proportion of all Rutgers students who prefer Hart.

b. What is the sample size that would be needed in estimating the above population proportion to within ± .006 with a confidence coefficient of 0.95.

4. The following data comes from an assay of insulin from blood sugar response of rabbits. Each rabbit was injected with 4 doses, two of the standard at 0.6 and 1.2 units ($S_1$ = A and $S_2$ = B) and two of the test preparation or unknown at presumably the same level. ($S_3$ = C, $S_4$ = D.)

Doses are labeled A, B, C, D

| Date April | Rabbit Number 1 | 2 | 3 | 4 | Total | Ave | sum of squared values/4 |
|---|---|---|---|---|---|---|---|
| 23 | B 24 | C 46 | D 34 | A 48 | 152 | 38 | 1538 |
| 25 | D 33 | A 58 | B 57 | C 60 | 208 | 52 | 2825.5 |
| 26 | A 57 | D 26 | C 60 | B 45 | 188 | 47 | 2387.5 |
| 27 | C 46 | B 34 | A 61 | D 47 | 188 | 47 | 2300.5 |
| Total | 160 | 164 | 212 | 200 | 736 | | |
| ave | 40 | 41 | 53 | 50 | ave=46 | | |
| sum of squared values/4 | 1757.4 | 1828 | 2931.5 | 2534.5 | | | |

Analyze the data. What do you conclude?

5. The following data come from 3 independent samples from three populations

| I | II | III |
|---|---|---|
| X | X | X |
| 15 | 17 | 6 |
| 18 | 22 | 9 |
| 12 | 5 | 12 |
| 9 | 15 | 11 |
| 10 | 12 | 11 |
| 12 | 20 | 8 |
| 20 | 14 | 13 |
| | 15 | 14 |
| | 20 | 7 |
| | 21 | |

a. Without assuming normality, test that the locations of the 3 populations are identical.

b. Determine which pairs of populations differ (use $\alpha = .05$) paying attention to the problem of multiple comparisons.